

Sirat Baweja

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EDUCATION

Trinity College Dublin MSc in Computer Science, Augmented and Virtual Reality Topics: Real-Time Rendering, Computer Graphics, Computer Vision, Machine Learning, Mathematics of Light and Sound, Motion Picture Engineering, Advanced Software Engineering, Extended Reality.	Sep 2025 - Sept 2026 (Expected) Dublin, Ireland
Amity University BTech in Computer Science and Engineering - CGPA 8.07/10.0, First Division. - Topics: Computer Graphics, ML Fundamentals, AI, Advanced Neural Networks, Deep Learning.	Aug 2017 - Jun 2021 Noida, India

WORK EXPERIENCE

Accenture <i>Senior Software Engineer</i> AVEVA PI, Python, React.js, Figma - Owned end-to-end delivery of a real-time industrial monitoring platform for the world’s largest diversified mining company, serving multiple sites across continents. - Built AVEVA PI-based data ingestion pipelines from industrial sensor networks, surfacing live operational data to TimescaleDB and Snowflake via REST APIs. - Designed dashboards in Figma and drove full implementation within PI Vision’s constrained legacy toolset, reducing client data interpretation time by 90%. - Automated bulk data management workflows with Python and JavaScript, saving 8+ hours per week. - Coordinated feature delivery across onshore/offshore teams and Australian client stakeholders across time zones.	Gurugram, India (Hybrid) Jun 2024 - Aug 2025
Accenture <i>Software Engineer</i> AVEVA PI, Python, .NET, RHEL - Built real-time data pipelines for critical iron ore mining operations at a global mining conglomerate using AVEVA PI over an 18-month engagement. - Structured time-series sensor data with contextual hierarchies and event frames for control systems engineers. - Administered TrendMiner, a RHEL-based big data analytics platform, across Windows and Linux servers with 99% uptime.	Gurugram, India (Remote) Sep 2021 - Jun 2024
koodos <i>Product Engineering Intern</i> Node.js, React (Next.js), Figma - Selected for the Koodos Collective, a highly selective group of engineers and creatives designing product experiments around creative expression. - Built “Caption Carnival”, an open-source Discord bot for captioning images and generating memes, fronted by mascot character “Capt. Shon.” - Collaborated with MIT Media Lab members in sprint sessions to prototype emerging tech applications.	Remote Oct 2020 - Mar 2021

PROJECTS

Real-Time VXPG (Voxel Path Guiding) C++, OpenGL 4.3, GLSL, compute shaders - From-scratch derived implementation of a SIGGRAPH 2024 real-time path guiding paper - ReSTIR GI and temporal denoising in OpenGL 4.3 with compute shaders (demo, writeup).	🔗 code
Local Research LLM (RAG Assistant) Python, FastAPI, PostgreSQL/pgvector, llama.cpp - Built an ingestion pipeline (PDF parsing, heading-based chunking) that embeds and stores dissertation research papers in a self-hosted PostgreSQL/pgvector store, synced with Zotero. - Implemented hybrid retrieval combining pgvector similarity search with PostgreSQL full-text search and cross-encoder reranking, served via FastAPI with local inference through llama.cpp.	🔗 code
Single-Image Stylized Depth Renderer C++, OpenGL, PyTorch, ONNX Runtime, OpenCV - Built a monocular depth estimation pipeline (Depth Anything V2, PyTorch, exported to ONNX Runtime). - Combined it with a 4-mode NPR cel-shading pipeline in OpenCV and OpenGL.	🔗 code
Procedural Forest Terrain and Foliage Renderer C++17, OpenGL 3.3, GLSL, WebGPU, TypeScript - TCD coursework project featuring procedural terrain, 300k+ GPU-instanced foliage, and a cel-shaded character with LOD. - Created a web version ported to WebGPU/TypeScript (live demo, writeup).	🎬 demo
VR Art Gallery (Project X) C#, Unity 6, Supabase, GitHub Actions - Co-built a multiplayer VR art creation and exhibition platform over two semesters. - Implemented JWT authentication with row-level security in Supabase for role-based access control. - Built a storage pipeline with automatic thumbnailing and CI/CD via GitHub Actions.	🔗 code

TECHNICAL SKILLS

- Languages:** Python, C++, C#, JavaScript, TypeScript.
- Graphics & Rendering:** OpenGL, GLSL, WebGPU, WebGL, Three.js, Unity 6, Unreal Engine 5.
- Web & Backend:** React.js, Next.js, Node.js, FastAPI, .NET.
- Data & Cloud:** PostgreSQL, pgvector, TimescaleDB, Snowflake, Supabase, AWS, Azure, Docker, Kubernetes.
- ML & Applied AI:** PyTorch, ONNX Runtime, OpenCV, Scikit-learn, llama.cpp.
- Tools:** Git, GitHub Actions, GitHub Copilot, Figma, Claude Code.